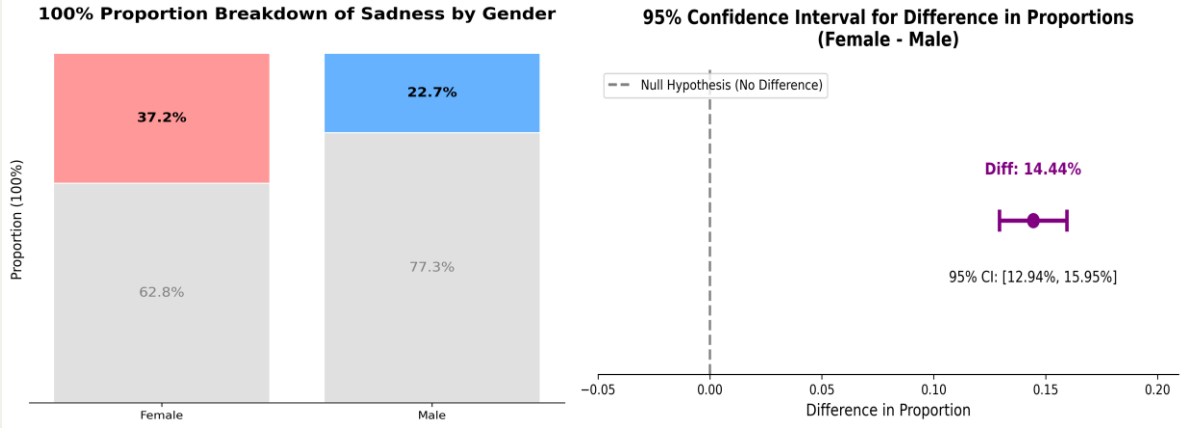


1 Basic EDA & Statistical Inference



Gender & Sadness Distribution

Female Sad or Hopeless Rate: 37.18% (2,580 / 6,940)

Male Sad or Hopeless Rate: 22.73% (1,567 / 6,893)

EDA Observation → Female students show a visibly higher proportion of feeling sad or hopeless than males.

Two-Proportion Z-Test Results

Total Sample Size: $n = 13,833$

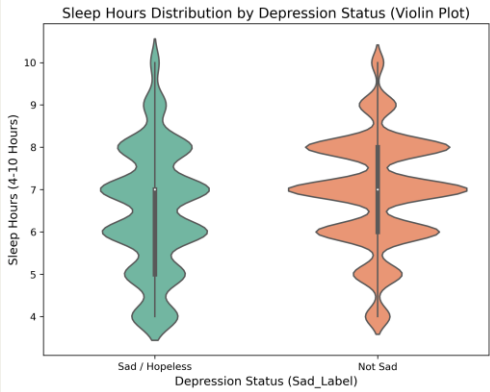
Sample Estimate : +0.1444 (+14.44%)

95% Confidence Interval: [0.1294, 0.1595] (★ Does not contain 0)

Z-statistic: 18.5374 | p-value: < 0.001

Decision: Reject H_0 (Highly significant difference in sad/hopeless rates between genders)

2 Sad or Hopeless Status vs. Sleep Duration



Sleep Distribution by Mental Health Status

Not Sad Group:

Sleep hours distribution peaks clearly around 7.0 hours. Shows a more stable, symmetric concentration within the recommended sleep range.

Sad / Hopeless Group:

The entire sleep distribution exhibits a noticeable downward shift.

Features a wider spread in the lower tails, indicating a higher prevalence of sleep deprivation.

EDA Observation:

Students experiencing sad or hopeless feelings tend to have their sleep duration significantly compressed.

3 Extension Statistical Inference

Welch's Two-Sample t-test on Sleep Hours

Methodology: Welch's t-test ($n = 12,106$)

Robust approach without assuming equal variances between groups.

Group Comparisons:

Not Sad Group Mean: ~7.00 hours

Sad / Hopeless Group Mean: ~6.50 hours

Statistical Estimations:

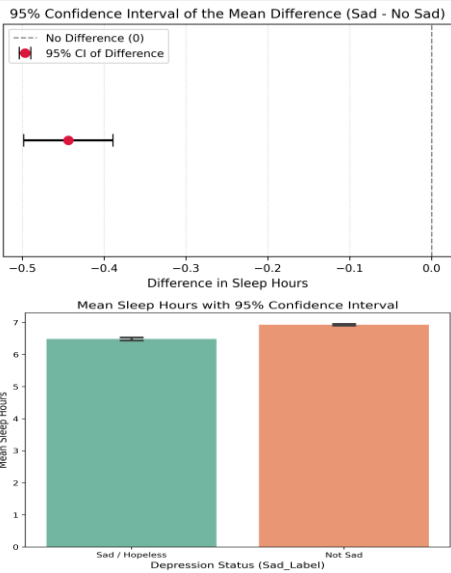
Mean Difference (Sad - Not Sad): -0.44 hours (~26 minutes less sleep)

95% Confidence Interval: [-0.49, -0.39] (★ Completely negative, avoids 0)

Hypothesis Testing:

p-value: < 0.001 (Highly significant)

Decision: Reject H_0 (Strong evidence that sad or hopeless feelings are linked to shortened sleep).



4 Conclusion

Gender Gap: Female high school students face a 14.44% higher risk of experiencing sad or hopeless feelings compared to males.

Sleep Deprivation: Psychological distress is heavily associated with physical sleep restriction, compressing sleep by nearly 26 minutes daily.

Observation Limit: This supports the association between emotional distress and sleep loss; however, as an observational study, causal relationships cannot be concluded.